Shravan S

cs23btech11053@iith.ac.in | shravans7654@gmail.com | +91 9940318559 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.85
Kendriya Vidyalaya IIT Powai	Class XII (CBSE)	2023	94.80%
Kendriya Vidyalaya IIT Powai	Class X (CBSE)	2021	99.60%

WORK EXPERIENCE

Flipkart Internet Private Limited - On Call Agent Automation Project

- **Automated** internal debugging workflows using **Python**, cutting **investigation time** by 20%
- Engineered **state-machine orchestrator** with human-in-the-loop pause/resume for 12 issue types
- **Prevented** injection attacks via regex-based **input validation** and command substitution rejection
- **Built** intuitive Streamlit UI, enabling on-call engineers to investigate without CLI access

EXTRACURRICULAR

Optiver - **Future Focus**

- Got Selected for Optiver's Future Focus **Trading and Research** Program
- Participated in various **Market Making**, **Guesstimates** and **Algorithmic Trading** Events

PROJECTS

GPUParallelizedPrefixSum | C++, CUDA

- **Achieved** a **458x speedup** over Baseline on **1M** data elements with a multi-block **CUDA Prefix Sum kernel**
- **Scalably parallelized** workload achieving **43.8x speedup** over **single-block execution** using 3-Phase scanning
- **Maximized** GPU Streaming Multiprocessor **occupancy** by distributing 970+ data chunks **concurrently**
- **Utilized shared memory** and **block synchronization** primitives to eliminate **global memory** read/write **latency**

UninitSanitizer | C++, LLVM IR, CMake

- **Engineered** a custom optimization pass within the LLVM framework to **Detect Uninitialized Variables**.
- **Implemented** instruction-level **def/use tracking** over a dominator-tree-ordered traversal of the CFG
- **Designed root-cause reporting** to suppress cascading duplicate warnings from one propagation chain

ParallelDistanceEngine | C++, CUDA

- **Developed** a custom Kernel with a **2.7x speedup** over Baseline for finding points within a certain Radius
- **Extracted** performance through memory coalescing and utilization of constant memory
- **Handled race conditions** through utilization of atomic operations to ensure correctness

Red Wine Quality Data Analysis | Python, Scikit-Learn, NumPy

- **Trained** and **evaluated** 5 models--including Random Forest, Multi-Layer Perceptrons, etc. **across train-test splits**
- **Engineered** a tuned **Random Forest** regressor that achieved the top out-of-sample accuracy
- Computed **Partial Dependence Plots** (PDP) across 11 features to explain non-linear feature interactions

RISC-V Assembler (Computer Architecture, CS2323) | C

- **Engineered** a two-pass assembler in C to translate RISC-V assembly instructions into binary machine code
- **Implemented** parsing with structured memory buffers to categorize instruction types and register operands
- **Designed** robust error-handling mechanisms to validate syntax and handle malformed inputs}

Stock market simulation game | Python, Streamlit, NumPy, MySQL

- Fast-paced, event-based **trading game**
- **Real-time handling** of P&L, positions, and price movements

RELEVANT COURSES

Computer Science : Discrete Mathematics, Data Structures, Algorithms, Computer Architecture, Database Management, Operating Systems, Theory of Computation, Compilers, Computer Networks

Mathematics : Calculus, Probability and Random Variables, Linear Algebra, Transform Techniques, Statistics

Machine Learning : Foundations of Machine Learning, Deep Learning

SKILLS

Languages : C++, Python, C, Java, JavaScript, SQL, Assembly (RISC-V)

Technologies : Streamlit, NumPy, Scikit-Learn, Matplotlib, Git, CUDA

POSITIONS OF RESPONSIBILITY

Coordinator, Finance and Consulting Club 2024-25

Buddy, Sunshine 2023-25

Teaching Assistant for **Compilers-II** (CS3423) July - Nov'26

CERTIFICATIONS

Global Consumer Intelligence Course, 2025

- Organized by the **Matsuo-Iwasawa Laboratory**, Graduate School of Engineering, **The University of Tokyo**
- Covered the fundamental theories and implementations of **data science**